Task 1: Open your Linux terminal and use the ps command to display all currently running processes for your user. Save the output to a file named my_processes.txt.

Ans :-

ps -u $USER > my_processes.txt

Verify the file:

cat my_processes.txt

Task 2: Start the Spotify client (or any music player app available on your Linux system), then use the top command to find its process ID (PID) and note it down.

Hint: Look for the process name in the top list or use pgrep if needed.

Ans :-

If Spotify is installed:

spotify &

If Spotify is not installed, you can use another media player such as VLC:

vlc &

Open another terminal and run:

top

Look for the process name (spotify or vlc) and note the **PID**.

Alternatively, use:

For Spotify:

pgrep spotify

For VLC:

pgrep vlc

Example output:

3245

PID = **3245**

Task 3: Use the kill command to terminate the process you found in the previous step. Verify that the process has stopped by running ps or top again.

Ans :-

Replace **3245** with your actual PID.

kill 3245

Verify it has stopped:

ps -u $USER

or

pgrep spotify

If nothing is returned, the process has been terminated.

If the process refuses to stop:

kill -9 3245

Task 4: Run a long-running command like ping google.com in the background using &. Then, bring it to the foreground and stop it using Ctrl+Z and the fg command.

Hint: Use the jobs command to see background jobs.

Ans :-

Start ping in the background:

ping google.com &

Check background jobs:

jobs

Bring it to the foreground:

fg %1

While it is running, press:

Ctrl + Z

This suspends the process.

Check jobs again:

jobs

You should see something similar to:

[1]+ Stopped ping google.com

Task 5: Suppose you accidentally started a heavy process that is slowing down your system (e.g., a big file copy or a simulation). Use the nice and renice commands to lower its priority and explain in one line how this helps system performance.

Ans :-

Start a low-priority process:

nice -n 10 ping google.com > /dev/null &

Find its PID:

pgrep ping

Further lower its priority:

renice 15 -p <PID>

Example:

renice 15 -p 4128

Verify the new priority:

ps -o pid,ni,cmd -p 4128

Example output:

```
  PID  NI CMD
 4128  15 ping google.com
```

One-line explanation:

Increasing the nice value lowers the process priority so it uses less CPU time, allowing more important processes to run more smoothly.